An Iterative Classification and Semantic Segmentation Network for Old Landslide Detection Using High-Resolution Remote Sensing Images

Zili Lu, Yuexing Peng, *Member, IEEE*, Wei Li, *Senior Member, IEEE*, Junchuan Yu, Daqing Ge, Lingyi Han, and Wei Xiang, *Senior Member, IEEE*

***Abstract*— The geological characteristics of old landslides can provide crucial information for the task of landslide protection. However, detecting old landslides from high-resolution remote sensing images (HRSIs) is of great challenge due to their partially or strongly transformed morphology over a long time and thus the limited difference with their surroundings. Additionally, small-sized datasets can restrict in-depth learning. To address these challenges, this article proposes a new iterative classification and semantic segmentation network (ICSSN), which can significantly improve both object-level and pixel-level classification performance by iteratively upgrading the feature extraction module shared by the object classification and semantic segmentation networks. To improve the detection performance on small-sized datasets, object-level contrastive learning is employed in the object classification network featuring a siamese network to realize global features extraction, and a subobject-level contrastive learning (SOCL) method is designed in the semantic segmentation network to efficiently extract salient features from boundaries of landslides. An iterative training strategy is also proposed to fuse features in the semantic space, further improving both the object-level and pixel-level classification performances. The proposed ICSSN is evaluated on a real-world landslide dataset, and experimental results show that it greatly improves both the classification and segmentation accuracy of old landslides. For the semantic segmentation task, compared to the baseline, the F1 score increases from 0.5054 to 0.5448, the mean intersection over union (mIoU) improves from 0.6405 to 0.6610, the landslide IoU grows from 0.3381 to 0.3743, the pixel accuracy (PA) is improved from 0.945 to 0.949, and the object-level detection accuracy of old landslides surges from 0.55 to 0.90. For the object classification task, the F1 score increases from 0.8846 to 0.9230, and the accuracy score is up from 0.8375 to 0.8875.**

***Index Terms*— Contrastive learning, landslide detection, multitask learning, semantic segmentation.**

I. Introduction

LANDSLIDES are natural disasters that occur frequently and can cause huge damages, such as casualties, building collapse, and farmland destruction, particularly to human life [1]. Therefore, it is crucial to analyze the statistics of existing landslides for accurate prediction of potential landslides to prevent loss of life and property. Large-scale geological data can efficiently help us understand the characteristics of landslides [2]. However, traditional methods of obtaining landslide geological data, such as field observation and mapping, are intensively time-consuming and thus cannot be applied in wide-area geologic surveys [3], [4]. Fortunately, with the rapid advancement of remote sensing technology, massive amounts of data have been accumulated and widely used for remote sensing tasks, including land utilization [5], environment monitoring [6], agriculture [7], forestry [8], and city planning [9]. The same trend has been observed in landslide detection, where remote sensing data is utilized for potential landslide detection and impact assessment [10].

For landslide detection using remote sensing data, manual visual interpretation is still the preliminary method, where specialists analyze the geomorphic manifestations of the disaster area and the landslide generation mechanism process then circle the landslides [11]. This kind of time-consuming and labor-intensive method is short of automation, intelligence, and robustness, as it heavily relies on the expertise of specialists [12], [13].

To improve the efficiency of landslide detection, a multitude of machine learning-based methods have been proposed. In [14], [15], and [16], methods are proposed to classify each pixel directly based on the spectral information of a high-resolution image. Alternatively, in [17], [18], [19], [20], and [21], objects with similar features are classified using spectral, spatial, hierarchical, and other features to extract landslide information. Although these machine learning methods initially achieve a wide range of automatic landslide detection, which greatly saves time compared to the manual visual interpretation method, their detection performance depends on manually selected features or feature engineering.

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Furthermore, they exhibit unsatisfactory generalization performance, which further restricts their applications over wide areas [22], [23].

With the explosive growth of convolution neural network-based deep learning methods, semantic segmentation models gradually become mainstream in the field of landslide detection. From region based convolutional neural network (R-CNN) [24], [25], [26] and fully convolutional network (FCN) [27], through to the DeepLab series [28], semantic segmentation models have demonstrated excellent feature extraction and prediction capabilities in various fields. For automatic landslide detection, many advanced semantic segmentation networks have been proposed and validated to offer significant improvements in efficiency and accuracy of large-scale detection [29], [30], [31], [32], [33], [34], [35]. Based on semantic segmentation, some change detection-based methods are also proposed to detect landslides. By comparing the surface images of different times in the same region, the surface change is detected and determined whether it is caused by landslide [52], [53], [54], [55], [56].

However, existing semantic segmentation model-based landslide detection studies have two problems. First, most of them aim at the visually salient new landslides which have clear boundaries and distinct differences in color with the background [31], [32], [38]. These models work poorly for old landslide detection. Second, generally, multimodal data can provide more information for landslide detection, and then multimodality fusion-based methods can improve the detection performance when compared with high-resolution remote sensing image (HRSI)-based methods. However, they greatly increase the demand on the dataset, which further worsens the small-sized dataset problem.

In this article, a novel iterative classification and semantic segmentation network (ICSSN) is proposed to detect visually blurred old landslides from a small-sized HRSI dataset. The proposed ICSSN utilizes a multiple-task learning-based iterative optimization method to enhance both object-level and pixel-level classification accuracy by iteratively training the feature extraction module shared by the object classification and semantic segmentation networks. Moreover, a subobject-level contrastive learning (SOCL) paradigm is developed to extract salient features from the back and side walls of landslides, which contribute most to reliable landslide detection. The iterative optimization and SOCL can fully utilize small-sized datasets and distill semantic features more efficiently.

The main contributions of this article are as follows.

1) The ICSSN is proposed for visually blurred old landslide detection using HRSI data. The ICSSN greatly improves the reliability of both pixel-level and object-level classification by iteratively optimizing the weight-sharing feature extraction networks shared by the object classification and semantic segmentation tasks. The highly reliable object classification network can guide the feature extraction of a highly complex semantic segmentation network, while the semantic segmentation network can supplement the semantic features for object classification, thus improving both pixel-level and object-level classification performance.
2) The SOCL method is developed in a semantic segmentation network for high-efficiency salient feature extraction. Motivated by the fact that side and back walls contribute most to old landslide detection, supervised contrastive learning is employed to extract salient features, and the boundary information is utilized to construct subobject-level positive samples, which is verified to be capable of distilling semantic features much more efficiently.
3) Extensive experiments are carried out to validate the efficacy of the proposed ICSSN. Our experimental results show that both the pixel-level and object-level classification performances are markedly improved, while the ablation study shows the iterative optimization strategy and the SOCL play important roles in improving the detection performance of our proposed ICSSN.

II. Related Work

The CNN-based deep learning methods can effectively extract semantic features and thus have been widely applied to detecting landslides.

Semantic segmentation models have attracted the most attention in landslide detection from HRSI data. In [30], a siamese convolutional neural network framework is proposed, where a generative adversarial network (GAN) is used to suppress the temporal difference between two HRSIs in different time phases, and a siamese network is employed for pixel-level change detection. A transformer network is introduced in [39] to enlarge the receptive field, while overlapping patch introduced to ascertain the relationship between the associated patches, and then feature extraction efficiency is improved through the use of a simple multi-layer perceptron (MLP) decoder.

Although many deep learning-based works have been proposed, their research efforts focus mainly on the detection of visually distinct landslides whose optical features are significantly different from the background, with little attention given to visually blurred old landslides. In a recent study [40], several mainstream semantic segmentation models are evaluated for old landslide detection and achieve the best F1 score of 0.417. In our previous work [41], we design the feature-fusion based semantic segmentation network (FFS-Net) model to extract semantic features from HRSI and digital elevation model (DEM) data, and then fuse the heterogeneous features in the semantic space to considerably enhance the landslide detection performance than the DeepLabv3+ scheme. DEM data can present more useful information for landslide detection, but also place a much higher requirement on the training dataset [42], [43], [44].

However, there remain big challenges for reliable old landslide recognition using only HRSI data, due to the following reasons.

1) *Visual Blur Problem of Old Landslides:* As old landslides normally remain inactive for a long time, their morphology features have been partially or strongly transformed by natural or human activities [36]. As a result, their color, tone, texture, and shadow features become gradually blurred, and their differences with surrounding slopes are severely reduced, which makes

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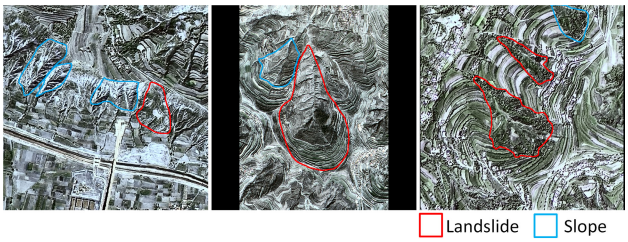


Fig. 1. Three old landslide samples in the Loess Plateau dataset to show the difficulty of recognizing old landslides from HRSI data.

reliable detection so difficult that even a geological expert may miss old landslides. We illustrate the visual blur between four landslides and their nearby slopes in Fig. 1, where the red and blue curves mark the landslide and slope, respectively.

2) *Small-Sized Datasets:* The reliable detection of visually blurred old landslides demands sufficient and well-labeled samples. However, the production of a high-quality landslide dataset from HRSI data is highly time- and labor-consuming, which requires significant professional expertise, leading to a scarcity of open large-scale datasets. Consequently, the size of available samples is far from enough to train high-complexity semantic segmentation models [37].

This article focuses on the challenging task of visually blurred old landslide detection using limited HRSI data. To address the problem caused by small-sized datasets, contrastive learning is exploited thanks to its effectiveness in feature extraction. Previous works have employed contrastive learning for both object classification and semantic segmentation. In [45], every instance and its data augmentation samples constitute the positive samples, while all the other samples constitute the negative ones. By comparing the similarities and differences of the contrastive samples, semantic features are extracted by a CNN and nonparameters softmax. A memory bank is introduced to save the features and the noise contrastive estimation (NCE) loss function is employed to calculate loss and perform back propagation. Finally, a compact feature space and a good classification accuracy are obtained. He et al. [46] construct a large and consistent dynamic dictionary with queuing and moving average encoders to facilitate unsupervised contrastive learning. The contrastive loss is further extended in [47] by designing a full-supervised contrastive loss so that it can effectively utilize label information. In [48], a pixelwise contrastive algorithm is proposed for semantic segmentation in the fully supervised setting. Semantic segmentation is improved by the contrastive learning strategy, and the contrastive loss is computed for each pixel during the training process to assist in optimizing the feature space. Building on these works, we improve the recognition reliability of old landslide from the following two aspects.

1) Inspiring by the expert knowledge from artificial visual interpretation approaches, the salient features for reliable old landslide detection mainly include: a) the height drop characteristics between landslide and background, especially at the location of the back wall and the side wall and b) the tongue or horseshoe shape of the deposits [36]. In this article, we designed an SOCL method to extract salient features from landslide side walls, back walls, or deposits, which induces the semantic segmentation model to focus more on these parts that contribute the most to the recognition results and achieve more reliable semantic feature extraction.
2) Under the principle of model collaboration for multitask learning [49], we designed an iterative optimizing method for feature extractor shared by object classification and semantic segmentation tasks. By sharing the feature extraction network, the global features of the landslide are extracted by the object classification network aided by OCL. Then the semantic segmentation network is instructed to extract locally salient features through SOCL, which in turn augments the object classification network with more comprehensive semantic features through iteration. Ultimately an overall improvement in target-level and pixel-level classification performance is achieved, which is verified by comparative and ablation experiments.

III. Proposed Method

We design a trainable network group and formulate a matched training algorithm to realize automatic old landslide detection. In what follows we first specify the overall architecture, details of each network of this network group, and finally the model training strategy.

A. Overall Structure

As shown in Fig. 2, we partition the ICSSN into two separate branches, namely the upper branch for object classification and the lower branch for semantic segmentation. Both branches consist of an encoder and a classifier (decoder), which extract semantic features from RGB images and yield object or pixel types, respectively. Through alternate training and iterative optimization, the upper branch guides the global feature extraction of the semantic segmentation network, while the lower branch extracts more abstract and comprehensive semantic features to improve the classification accuracy of the upper branch.

B. OCL Augmented Object Classification Network

We designed a classification network with the Siamese structure to detect whether samples contain landslides by OCL. As shown in Fig. 2 (top half), the classification network consists of a feature extraction module and a classifier module.

1) Encoder: The feature extraction module adopts a weight-sharing Siamese network architecture, which is composed of two encoders. Inspired by the classical DeepLabv3+ network structure, a squeeze-and-excitation (SE) attention module and an encoder-to-decoder skip connection structure are introduced at the encoder side. Each encoder consists of the backbone (ResNet101), atrous spatial pyramid pooling (ASPP), and SE modules for the extraction of high-dimensional features from the input RGB pictures. These modules are detailed as follows.

1) The ResNet101 is employed as the backbone for feature extraction attributed to its proven effectiveness in

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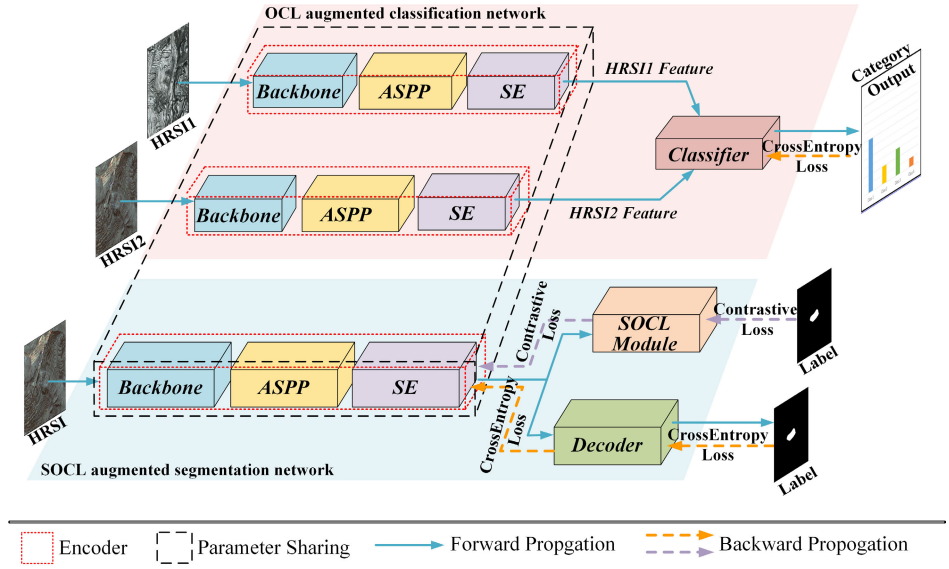


Fig. 2. Architecture of the ICSSN, including a classification network augmented by object-level contrastive learning and a semantic segmentation network augmented by SOCL. There are three encoders in the ICSSN architecture all of which share the same parameters.

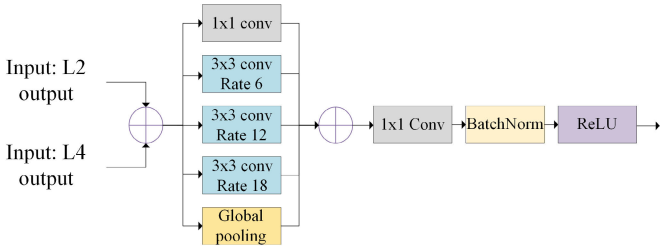


Fig. 3. ASPP structure.

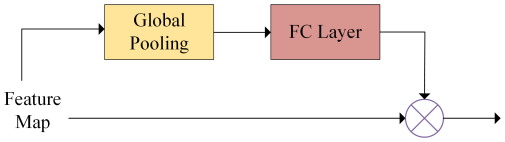


Fig. 4. SE structure.

feature extraction. It comprises four residual blocks, each of which has a different resolution of feature maps. After cross-validation, we found that concatenating the feature maps of layers 2 and 4 produces a final output that balances between abstract features and detailed information;

2) The ASPP module is inserted for multiscale feature fusion. By using the dilation convolution with different receptive fields to sample the feature maps, the encoder can effectively take into account both small-scale detailed texture information and large-scale overall morphology information simultaneously. Fig. 3 shows the structure of ASPP in detailed.
3) As shown in Fig. 4, SE module adaptively weights all channels in the feature map, allowing important channels to have greater weights and yielding a substantial enhancement in the capability to extract features.

2) Classifier: As depicted in Fig. 5, the classifier module consists of a global maximum pooling layer and two fully-connected layers, which maps and weights the extracted

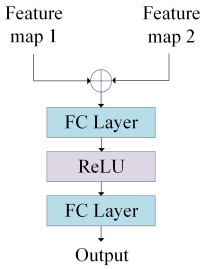


Fig. 5. Object-level contrastive-learning-based classification network.

features to obtain the prediction probability for each category. The category with the highest prediction probability is yielded.

Compared with classic classification networks, we employ a weight-sharing Siamese network to improve the feature extraction capability and then concatenate the extracted global semantic features. This design offers the following two advantages. First, it allows the encoder to process two images simultaneously and concatenate the features, providing more features for the network to learn. Second, two input samples can form four distinct combinations for the classification prediction, i.e., landslide-landslide, landslide-slope, slope-landslide, and slope-slope, so that the network can learn the semantic features more efficiently by comparing the similarity and difference between the two samples. Specifically, when two samples are of the same category (positive-positive pair or negative-negative pair), the encoder learns the common features of this category. When the input samples fall into different categories (positive-negative pair or negative-positive pair), different features are learned for the corresponding categories.

C. SOCL Augmented Semantic Segmentation Network

As depicted in Fig. 2 (bottom half), the semantic segmentation network is composed of three components, namely an encoder, a SOCL module, and a decoder. The SOCL module

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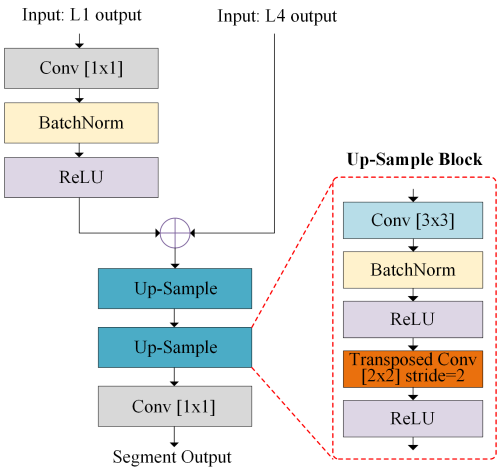


Fig. 6. Decoder structure in the semantic segmentation network.

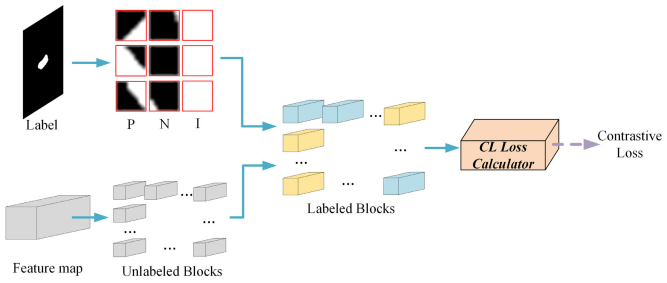


Fig. 7. SOCL module. In the SOCL labels, P represents the edge of a landslide, N represents the slope, and I represents the irrelevant class.

is a novel design that leverages a priori expert knowledge and supervised contrastive learning to enhance the capability of salient semantic feature extraction. In addition, the decoder uses a learnable reverse convolution layer instead of the traditional linear interpolation layer to improve the pixel-level prediction accuracy.

1) Encoder: The encoder of the segmentation network has the same structure as the classification network, so as to facilitate alternate training.

2) Decoder: Based on the feature maps, the decoder classifies each pixel. We resort to our proposed decoder in [41]. As shown in Fig. 6, two transposed convolution layers are stacked to recover the resolution and avoid information redundancy to the greatest extent. In addition, the dropout layer is employed to avoid over-fitting. The batch normalization layer is used to restrict the data fluctuation range, while the rectified linear unit (ReLU) activation function is employed to increase network sparsity and avoid gradient disappearance.

3) SOCL Module: Fig. 7 illustrates the SOCL module. The training of this module includes three steps, i.e., 1) using a frame of size 8×8 to filter the labels of the training images so as to obtain SOCL labels; 2) splitting the feature map in the spatial dimension, labeling it with SOCL labels and obtaining SOCL blocks; and 3) calculating the supervised contrastive loss by SOCL blocks.

For the SOCL blocks of size 8×8, some are identified as the boundary of a landslide and then selected as the positive SOCL samples when they contain type-1 pixels (i.e., pixels

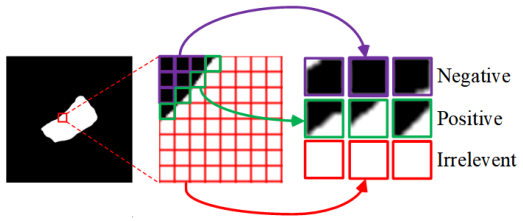


Fig. 8. Process of label filtering. Labels on the right are used in the SOCL module.

located in a landslide) with a number lying between 7 and 57. When an SOCL block contains less than seven type-1 pixels, it is regarded as background and treated as a negative SOCL sample. In particular, when an SOCL block contains more than 57 class-1 pixels, it is considered an internal part of a landslide and will be discarded due to its high similarity to slopes and thus a very limited contribution to the recognition of the landslide. This strategy is verified by the cross-validation experiment detailed in Section V. The selected positive, negative, and discarded SOCL samples are labeled as the 1, 0, and irrelevant classes, respectively. For each label $l \in \mathbb{R}^{H\times W\times 2}$, we obtain the corresponding label matrix $l' \in \mathbb{R}^{(H/8)\times(W/8)\times 3}$.

In the training phase, we normalize the feature map yielded by the encoder $f \in \mathbb{R}^{(H/8)\times(W/8)\times C}$ in the spatial domain, and this step does not change the dimension of f. It should be noted that f and l' are of the same spatial resolution and can be aligned.

Then we flatten both f and l' in the spatial domain and align features $f_i^{1\times 1\times C} \in f$ and labels $l_i^{'1\times 1\times 3} \in l'$ to form new subobject-level feature-label pairs, each of which represents an 8×8 block in original samples. After that, we place all the SOCL pairs together to obtain $(H/8) \times (W/8) \times$ batch_size pairs. Ignoring the pairs with labels of the irrelevant class, we randomly select a fixed number of pairs with classes 0 and 1 and then calculate their loss by the following supervised contrast learning loss function:

$$L_i^C = \frac{1}{|P_i|}\sum_{i^+\in P_i} -\log\frac{\exp\left(i\cdot i^+/\tau\right)}{\exp(i\cdot i^+/\tau)+\sum_{i^-\in N_i}\exp(i\cdot i^-/\tau)} \tag{1}$$

where P_i denotes the total number of positive SOCL samples, i^+ and i^- represent the positive and negative SOCL samples, respectively. Meanwhile, the following cross-entropy loss function is employed to measure the difference between the predicted and real labels:

$$L_i^{\mathrm{CE}} = -y_i\log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i) \tag{2}$$

where y_i indicates the label of pixel i, and $\hat{y}_i$ indicates the logical output of this pixel. Therefore, the loss function of the semantic segmentation network is the combination of the two losses in (1) and (2)

$$L^{\mathrm{Seg}} = \sum_i \left(L_i^{\mathrm{CE}} + \lambda L_i^C\right) \tag{3}$$

where λ is the preset weight. After cross validation, we found that the network is not sensitive to λ, so we set $\lambda = 1$ generally.

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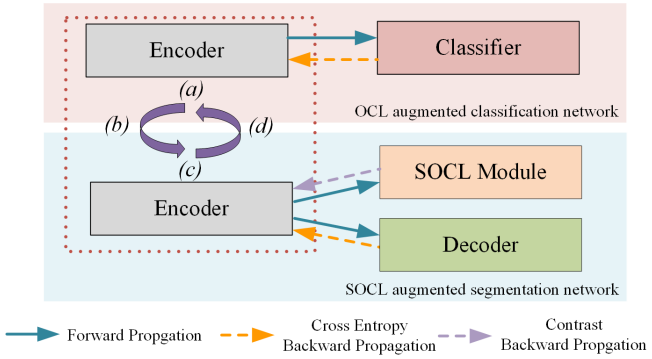


Fig. 9. Iterative training of the two networks, which share the same encoder. (a) Corresponds to steps 2, 7, and 8 in Algorithm 1, (b) corresponds to step 3, (c) corresponds to steps 4 and 5, and (d) corresponds to step 6.

D. Training Strategy

A collaborative model training strategy is designed to iteratively train the two branches, and then continuously update the shared encoder. The iterative training process is illustrated in Fig. 9.

The iterative training process alternates between the two networks. That is, the upper branch of the object classification network is trained until convergence, followed by the training of the lower branch of the semantic segmentation network until convergence. The whole training process is completed when both networks converge. During this alternation process, the encoder parameters in the previously converged network are used as the initial parameters for the latter network. The OCL augmented classification network is of high accuracy and can extract key features for object-level classification. Based on multitask learning, it can effectively provide guidance and iterative calibration for the task of feature extraction of the SOCL augmented semantic segmentation network through the iterative mechanism. Generally, the object classification network extracts more abstract and global features with greater reliability due to its simpler task, while the semantic segmentation network extracts more local and detailed features with lower reliability due to the much harder task of pixel-level classification. Through iterative training, the encoder learns both local detailed features and global abstract features so that the two networks can realize the object-level and pixel-level classification, respectively.

Before each iteration, warm-up training is performed. The encoder parameters are frozen, and the decoder (classifier) is trained until convergence. After the warm-up training, the encoder parameters are unfrozen, and joint training is performed alongside the decoder (classifier). The warm-up training prevents feature mismatch between the optimized encoder of the previous network and the decoder (classifier) of the last iteration of the latter network and helps verify the sharing feature space. The training process is listed in Algorithm 1.

Algorithm 1 Iterative Model Training

Initialize:

N_c: Classification network, containing $Encoder$ and $Classifier$

N_s: Segmentation network, containing $Encoder$ and $Decoder$.

1: **repeat**
2: Train N_c until convergence.
3: Load convergent $N_c(Encoder)$ parameters to $N_s(Encoder)$
4: Freeze $N_s(Encoder)$ and train $N_s(Decoder)$ until convergence.
5: Unfreeze $N_s(Encoder)$ and train N_s until convergence.
6: Load convergent $N_s(Encoder)$ parameters to $N_c(Encoder)$
7: Freeze $N_c(Encoder)$ and train $N_c(classifier)$ until convergence.
8: Unfreeze $N_c(Encoder)$
9: **until** Loss of N_c and N_s minimal.

IV. Experiments

In this section, the proposed ICSSN is evaluated ion a real-world dataset. All experiments are conducted using PyTorch on two Nvidia 3090 GPUs, a 12th Gen Intel[1] Core i9-12900K, and 128 GB of memory.

A. Experimental Dataset

The experiment zones are located in the northwest of China, which is in the transition zone between the western Qinling Mountains and the Longxi Loess Plateau. The soil parent material in this area is eluvial and accumulative, leading to soil with weak water permeability and erosion resistance. Sparse vegetation and heavy rainfall contribute to poor geological stability, which is conducive to the occurrence of landslides.

The landslides in the study area occurred long time ago. The shape, color, and texture characteristics of landslides are of close resemblance to the surrounding environment. Fig. 10 shows an old landslide sample which is hardly distinguishable from the background. The landslide boundaries are fuzzy and the back walls of the landslides are visually blurred. To make matters worse, the main bodies of some landslides are transformed into farmland or residential areas, making landslide detection much more challenging.

There are 304 landslide samples in this study area. The HRSI data have a resolution of 2 m/pixel. Fig. 11 shows some landslide samples in one country.

B. Data Preprocessing

The original HRSIs are large in size. To facilitate model training, the images are resized into uniform samples of size 512 × 512. All the images containing landslides are selected to create positive samples after annotation. Images with no landslides are randomly collected to form negative samples.

All the samples are divided into the training, validation, and test sets with a ratio of 6:2:2. Histogram equalization is applied to the entire dataset, and data augmentation techniques,

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Fig. 10. Landslide sample of the dataset, including the image and label.

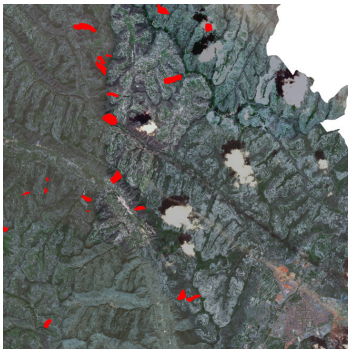


Fig. 11. HRSI landslide sample of one city. The red parts represent landslides.

TABLE I
DATASETS PARTITION

	Training	Validation	Test
Slide	1080	360	60
Non-slide	360	120	20

including horizontal flipping, vertical flipping, and rotation, are used on the training and validation sets. The number of samples in each set is listed in Table I.

C. Performance Metrics

Both object-level classification metrics and pixel-level semantic segmentation metrics are used. The object-level classification metrics include pixel accuracy (PA), precision, recall, and F1-score, while the pixel-level classification metrics include PA, precision, recall, F1-score, and mean intersection over union (mIoU), which are commonly employed

$$\text{accuracy} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{FP}+\text{FN}+\text{FP}} \tag{4}$$

$$\text{precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{5}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{6}$$

$$\text{Landslide-IoU} = \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \tag{7}$$

$$\text{mIoU} = \frac{1}{2} \times (\text{class } 1-\text{IoU} + \text{class } 0 - \text{IoU}) \tag{8}$$

$$\text{F1} = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \tag{9}$$

where TP, TN, FP, and FN represent the true positive, true negative, false positive, and false negative at the pixel level, respectively.

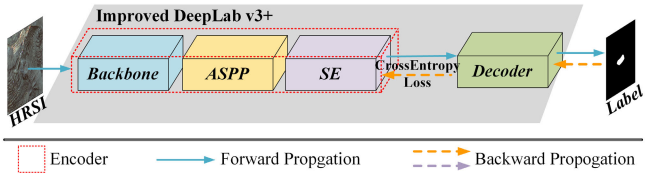


Fig. 12. Architecture of DeepLabv3+.

TABLE II
MODEL HYPERPARAMETERS

	Classification	Segmentation
Number of workers	4	4
Batchsize	4	4
Optimizer	SGD	SGD
Momentum	0.9	0.9
Initial learning rate	0.001	0.007
Weight decay	0.0005	0.0005
Learning rate decay strategy	cosine annealing	cosine annealing
Epoch	50	100

Additionally, an object-level classification metric can be obtained from the semantic segmentation metrics. Specifically, for a landslide sample, if more than 400 pixels within the landslide are accurately detected, it is considered a correctly classified sample. For a slope sample, if less than 100 pixels are incorrectly classified as belonging to a landslide, it is considered a correctly classified slope sample

$$\text{acc}_+ = \frac{1}{N_L}\sum_{i=1}^{N_L} I(C(\hat{P}_i \cap P_i) \geq 400) \tag{10}$$

$$\text{acc}_- = \frac{1}{N_S}\sum_{j=1}^{N_S} I(C(\hat{F}_j \cup F_j) \leq 100) \tag{11}$$

where N_L and N_S are the numbers of landslide and slope samples in the test set, respectively, $\hat{P}_i$ and $\hat{F}_j$ are the predicted labels of the landslide and slope samples, respectively. $I(x)$ is the index function whose value is 1 when x is true, or 0 when x is false. $C(x)$ is a function to count the number of pixels with the same label in two comparison pictures.

D. Reference Models

Although several landslide detection models [50] and heterogeneous information fusion models [42], [43], [44] have been proposed, unfortunately, none of them has yet published the source code of their models. As a result, we cannot reproduce their results for comparison purposes. Furthermore, we cannot directly compare our model with the heterogeneous information fusion models [42], [43], [44] for three additional reasons: 1) the target of these models are not landslides and 2) aerial images, DSM or SAR are used by these models, which are very different from HRSI data. Consequently, we have to evaluate the proposed model through comparative experiments with the baseline model and ablation experiments.

Four comparative experiments are designed: 1) improved Deeplabv3+ [41] as the baseline model, whose architecture is shown in Fig. 12; 2) segmentation network augmented by SOCL; 3) object classification network augmented by OCL; and 4) the proposed ICSSN. The hyperparameters of model

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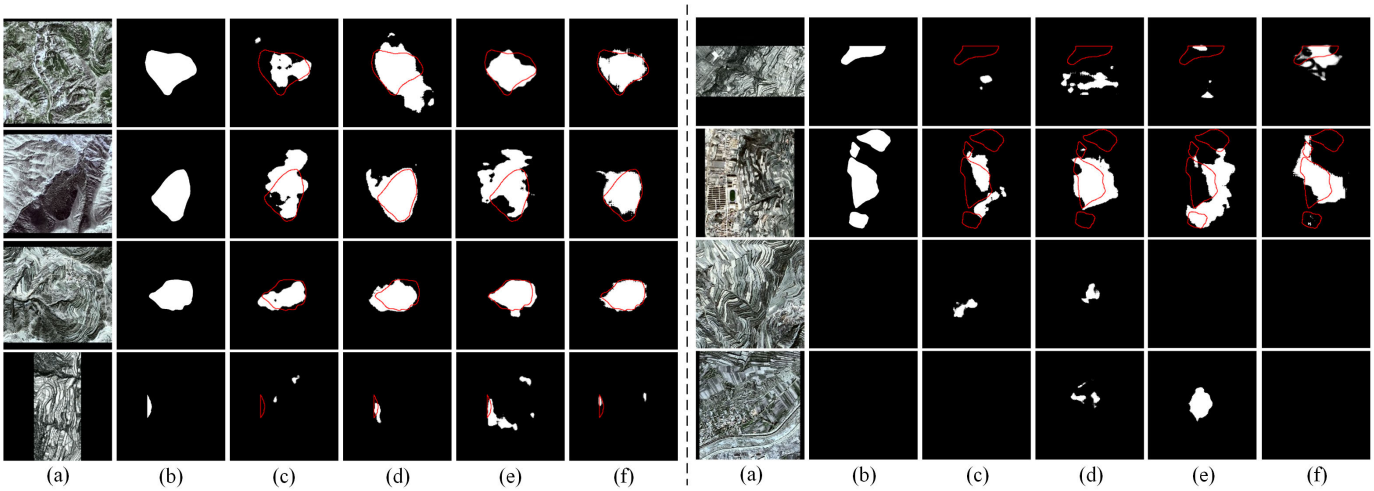


Fig. 13. Visual results of segmentation on old landslides. (a) RGB image. (b) Label. (c)–(f) Baseline, SOCL, OCL, and ICSSN.

TABLE III
PIXEL-LEVEL QUANTITATIVE EXPERIMENTAL RESULTS OF SEGMENTATION NETWORKS

Method	PA	Precision	Recall	Landslide-IoU	mIoU	F1
Baseline	0.9445	0.4226	0.6284	0.3381	0.6405	0.5054
SOCL	0.9406	0.5176	0.5621	0.3688	0.6536	0.5387
OCL	0.9431	0.4896	0.5911	0.3658	0.6535	0.5356
ICSSN	**0.9493**	0.4531	0.6832	**0.3743**	**0.6610**	**0.5448**

TABLE IV
OBJECT-LEVEL CLASSIFICATION EXPERIMENTAL RESULTS OF THE SEGMENTATION NETWORKS

Method	Landslide Acc	Slope Acc	Avg. Acc
Baseline	0.78	0.55	0.67
SOCL	0.78	0.55	0.67
OCL	0.78	0.65	0.72
ICSSN	0.78	**0.90**	**0.84**

training are the same for all the four comparison models, which are listed in Table II.

The SOCL model employs the cross entropy as the segmentation loss function, which is defined as follows:

$$\text{Loss} = -y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \tag{12}$$

where y_i indicates the ground truth and $\hat{y}_i$ denotes the predicted mask.

V. RESULTS AND DISCUSSION

In this section, we present experimental results with visualization as well as some detailed discussions on the obtained results.

A. Comparative Experiments

In this section, we will show and discuss the experimental results achieved by the four comparison models on the same old landslide dataset.

1) Semantic Segmentation Results: The pixel-level classification experimental results are listed in Table III, while the object-level classification results, which are derived from the segmentation network, are listed in Table IV. The following observation can be made from the results in these two tables.

1) The proposed ICSSN model achieves the best performance among all the comparison models. Specifically, compared to the baseline model, the F1 score increases from 0.5054 to 0.5448, the mIoU improves from 0.6405 to 0.6610, and especially the landslide-IoU is up from 0.3381 to 0.3743. Furthermore, the object-level accuracy of slope classification is significantly improved from 0.55 to 0.9. These improvements clearly demonstrate the advantage of the proposed model in both object-level and pixel-level classification of old landslides.
2) The effectiveness of the proposed SOCL module in improving boundary detection accuracy is demonstrated by the significant improvements in both pixel-level and object-level metrics when compared to the baseline model. Although the object-level accuracy of both classes remains the same as the baseline model, the SOCL network still achieves notable pixel-level improvements, with the F1 score increased from 0.5054 to 0.5387, the mIoU improved from 0.6405 to 0.6536, and the IoU of class 1 up from 0.3381 to 0.3688.
3) Regarding the object-level results of the OCL augmented segmentation network, slop accuracy increases from 0.55 to 0.65. As a result, the pixel-level performance is improved correspondingly. The F1 score increases from 0.5054 to 0.5356, the mIoU improves from 0.6405 to 0.6535, and the IoU of class 1 is up from 0.3381 to 0.3658.

Some experimental results are also visualized in Fig. 13. As can be seen from Fig. 13, the SOCL scheme predicts more accurate shapes, especially the boundary parts, when compared with the baseline scheme. The OCL scheme has fewer

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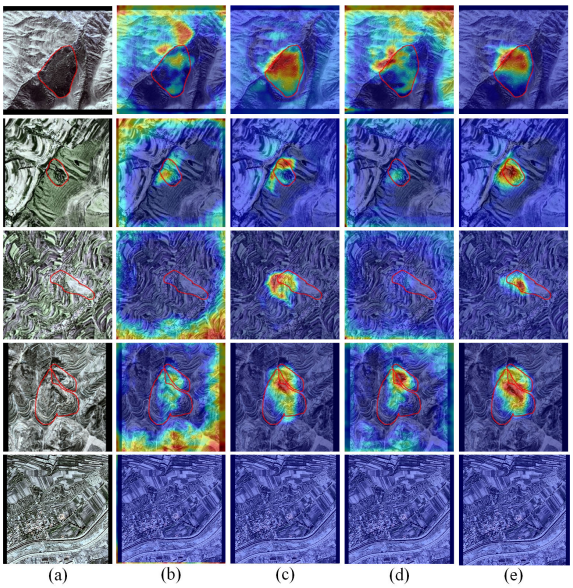


Fig. 14. Visual results by Grad-CAM of the segmentation networks. (a) Img. (b)–(e) Baseline, SOCL, OCL, and ICSSN. The first four samples are landslides with red lines indicating their boundaries, while the last one is a slope sample.

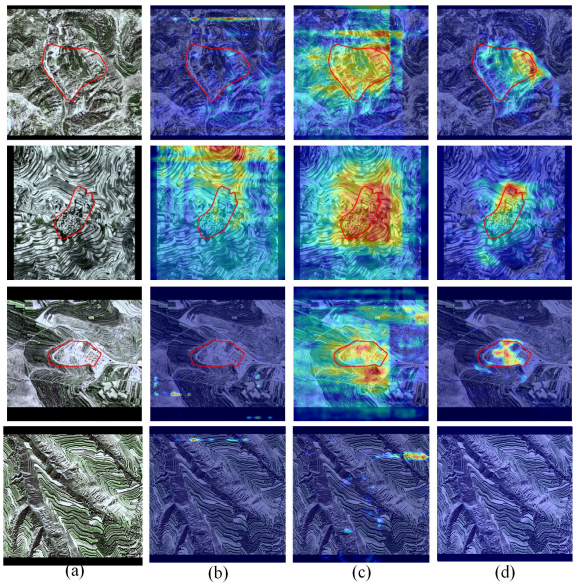


Fig. 15. Visual results by Grad-CAM of the classification networks. (a) Img. (b)–(d) No iteration, OCL, and ICSSN. The first three samples are landslides with red lines indicating their boundaries, while the last one is a slope sample.

TABLE V
QUANTITATIVE CLASSIFICATION EXPERIMENTAL RESULTS OF THE CLASSIFICATION NETWORKS

Method	Accuracy	Precision	Recall	F1
No iteration	0.8375	0.9607	0.8167	0.8846
OCL	0.8625	0.9455	0.8667	0.9044
ICSSN	**0.8875**	0.9473	0.9003	**0.9230**

TABLE VI
CROSS-VALIDATION OF THE SOCL BLOCKS SELECTION STRATEGY

Strategy	Landslide-IoU	mIoU	F1
Center	0.3427	0.6399	0.5105
Edge	**0.3688**	**0.6536**	**0.5387**
Hybrid	0.3606	0.6482	0.5300

misjudgments and omissions, while the ICSSN model, which fuses both the SOCL and OCL schemes, is most accurate in both object-level classification and pixel-level classification.

The heat map analysis can effectively validate the design of a model, and we have employed the popular gradient-weighted class activation mapping (Grad-CAM) algorithm [51] as our visualization method. The Grad-CAM method visualizes the contribution of each region of the input image to the predicted result by weighting the feature map and overlaying it with the original image. Fig. 14 visualizes the final layer of the encoder of our model. The heat maps demonstrate that the crucial areas in the heat map of the two schemes with the SOCL strategy are closer to the back and side walls, which verifies the effectiveness of the SOCL sample selection strategy and the SOCL module for detecting old landslides.

2) Object Classification Results: In both the OCL and ICSSN schemes, we introduced the classification networks. Table V lists the numerical results of these three classification schemes, i.e., no iteration, OCL object classification network, and ICSSN object classification network. Note that Tables IV and V both present object-level performances, but they are derived from different networks. Table V is derived from the classification network, while Table IV is derived from the segmentation network.

As can be seen from the result that the following holds.

1) The ICSSN achieves the best F1 score followed by the OCL, and both are better than the no iteration scheme. This shows that the segmentation network can provide effective guidance to the classification network and strengthen the feature extraction ability of the encoder by using the collaborative model training strategy. This guidance is still effective when the segmentation network has better abilities.
2) Fig. 15 visualizes the heat map of the classification network. It can be observed that the features extracted from the initial classification are not intuitive, and the prediction results are based on the entire image. Through iterative training with the segmentation network, the important area gradually shifts to the surroundings of the landslide, thereby improving the detection accuracy. Based on the above analysis, it is verified that the iterative training strategy is conducive to fusing the feature spaces of the classification network and the segmentation network, thus improving the performance of both branches.

B. Cross-Validation

In order to obtain the best performance, we optimize the design of the proposed model via cross-validation experiments.

1) SOCL Blocks Selection Strategy: We devise three strategies for selecting the SOCL blocks: 1) central strategy—the SOCL blocks are selected from the central parts of the landslides and they contain more than 57 class-1 pixels; 2) edge strategy—the SOCL blocks are chosen exclusively from the boundaries of the landslides and they contain class-1 pixels between 7 and 57; and 3) hybrid strategy—the SOCL blocks are randomly sampled from both the central and boundary parts of the landslides with equal probability.

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TABLE VII
CROSS-VALIDATION OF CLASSIFIER

Method	Accuracy	Precision	Recall	F1
Single-branch	0.8250	0.9432	0.8167	0.8750
Max Pool & 1×1 conv	0.7375	0.9149	0.7167	0.8037
Max Pool & $1*$FC	0.8000	0.9400	0.7833	0.8545
Avg Pool & $2*$FC	0.7125	0.8776	0.7167	0.7889
Max Pool & $2*$FC	**0.8375**	0.9607	0.8167	**0.8846**
Max Pool & $3*$FC	0.7500	0.9000	0.7500	0.8182

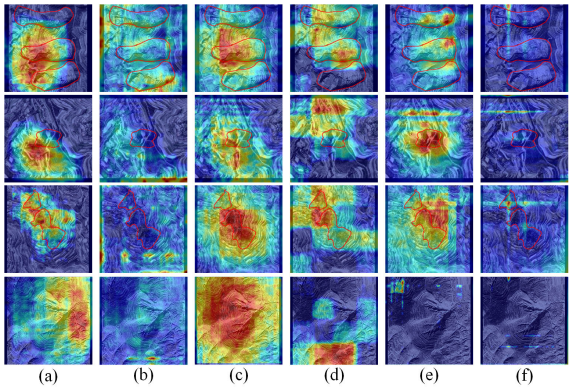


Fig. 16. Visual results by classifier cross-validation. (a)–(f) Correspond to six schemes in Table VII. The first three samples are landslides with red lines indicating their boundaries, while the last one is a slope sample.

TABLE VIII
QUANTITATIVE EXPERIMENTAL RESULTS OF THE CLASSIFICATION NETWORK DURING ITERATION

Method	Accuracy	Precision	Recall	F1
First round	0.8375	0.9607	0.8167	0.8846
Second warm-up	0.8250	0.9423	0.8167	0.8749
Second round	**0.8875**	**0.9473**	**0.9000**	**0.9230**
Third warm-up	0.8625	0.9623	0.8500	0.9027
Third round	0.8750	0.9629	0.8667	0.9122

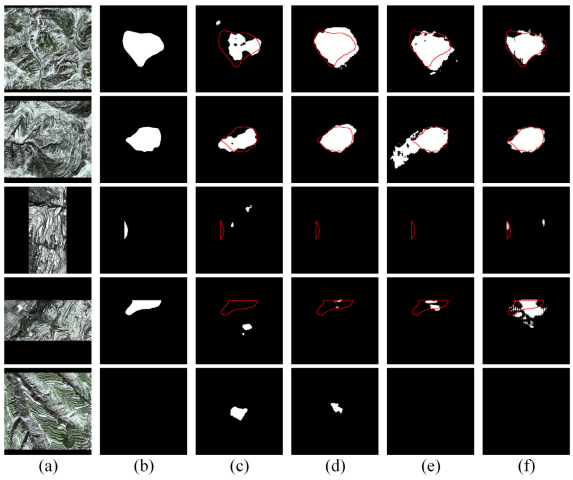


Fig. 17. Visual results by segmentation cross validation. (a) RGB img. (b) Label. (c) Baseline. (d) After first round. (e) After second round warm-up. (f) After second round. The first four samples are landslides with red lines indicating their boundaries, while the last one is a slope sample.

TABLE IX
QUANTITATIVE EXPERIMENTAL RESULTS OF THE SEGMENTATION NETWORK DURING ITERATION

	Landslide-IoU	mIoU	F1
First warm-up	0.2403	0.5880	0.3875
First round	0.3724	0.6567	0.5419
Second warm-up	0.3690	0.6551	0.5391
Second round	**0.3743**	**0.6610**	**0.5448**
Third warm-up	0.3652	0.6516	0.5350
Third round	0.3715	0.6566	0.5418

TABLE X
TRAINABLE PARAMETERS AND COMPLEXITY OF THE MODELS

	Params (M)	GFLOPs
Upper branch	62.92	128.59
Lower branch	63.68	150.03

The experimental results presented in Table VI indicate that the edge strategy performs the best, while the center strategy performs the worst. This observation confirms that the boundaries of a landslide contain more information relevant to landslide detection.

2) Classifier: We design several classification networks, including the single-branch and two-branch classification networks with five types of classifiers. The numerical and visual experimental results are listed in Table VII and Fig. 16, respectively, and both results show that the Max pool and $2*$FC schemes are preferred.

3) Iteration Process: The strategies of alternate training of two networks and iterative optimization of the encoder play an important role in the proposed model. We record the performance of the segmentation and classification networks during the model training in Tables VIII and IX and Fig. 17. As can be seen, the second-round performance of both networks is greatly improved compared with the first round, and the third-round performance is similar to the second round. This result is consistent with our design principle and verifies the effectiveness of the iterative training strategy.

Particularly, in the second warm-up stage of the segmentation network, the pixel-level prediction accuracy reaches a high level, demonstrating that the classification network and segmentation networks have achieved unification and sharing of the feature space to a certain extent by using the iterative training strategy.

C. Complexity Analysis

We also conduct a complexity analysis of the proposed model in terms of model parameters, GFLOPs, and runtime, where FLOPs stand for a billion floating-point operations per second. The results are listed in Table X, which are based on input HRSI pictures of dimension $512 \times 512 \times 3$. Moreover, the total time consumption is shown in Table XI.

Compared to the baseline model, the SOCL model consumes almost three times more time because of the extra calculation of the contrast loss function. Due to the introduction of the iterative training strategy, both the OCL network and ICSSN need to be trained several times, which leads to

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TABLE XI
TOTAL TIME CONSUMPTION

Method	Traing time (min)
Baseline	62.4
SOCL	192.8
OCL	380.5
ICSSN	674.1

increased training time. Taking into account the performance improvement, we believe this linear increase in computation is acceptable.

VI. CONCLUSION

This article proposed the ICSSN for old landslide detection using HRSI data, which is characterized by iterative optimization between the classification network and segmentation network to achieve effective extraction of global abstract information and local detailed information. In the proposed ICSSN, the classification network employs the OCL strategy, uses the double-branch parameter-sharing siamese network as the feature extractor, can simultaneously process a group of inputs, extract and fuse two input features, and realize the learning of the global features of a landslide with the aid of the classifier. In addition, the SOCL module is designed to assist in segmentation network training. This module frames and screens the features of key areas such as the side and back walls of a landslide, and uses the feature extractor the loss auxiliary training for improving the learning of the detailed features of the landslide. A collaborative model training strategy was designed, which involves alternate training of the two networks and iterative optimization of the shared encoder by both networks. The proposed model was evaluated sufficiently on a real-world old landslide dataset via comparative and cross-validation experiments. Extensive experimental results were presented to demonstrate that the proposed ICSSN is able to greatly improve the landslide detection performance at both object and pixel levels.

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